# A12.5 Reversible Reactions

Dynamic equilibrium • Le Chatelier’s principle • concentration • temperature • pressure

# Learning outcomes

- Define dynamic equilibrium and state the conditions required for it.
- Explain that forward and reverse reactions continue at equal rates at equilibrium.
- Use Le Chatelier’s principle to predict the effect of concentration changes.
- Predict the effect of temperature changes using exothermic and endothermic directions.
- Predict the effect of pressure changes for gaseous equilibria using numbers of gas particles/moles.

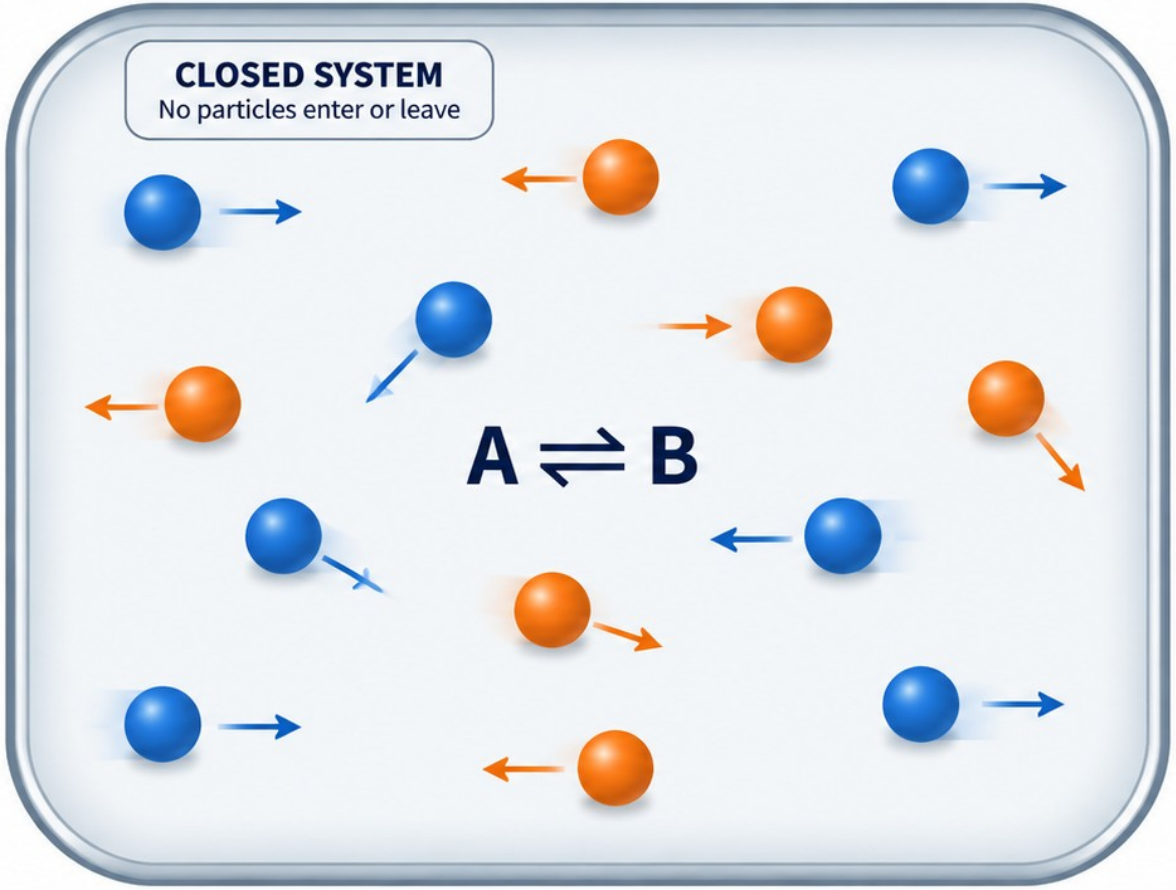
# Reversible reactions

- A reversible reaction can proceed in both directions.
- The forward reaction forms products; the reverse reaction reforms reactants.
- Use the reversible-reaction symbol ⇌ rather than a single arrow.
- Example model: A + B ⇌ C + D.
- Reversible does not automatically mean the system is at equilibrium.

# A12.5.1 Definition of dynamic equilibrium

- Dynamic equilibrium is established in a reversible reaction in a closed system when the forward and reverse reactions occur at equal rates.
- Both reactions continue: the equilibrium is dynamic, not static.
- Because the rates are equal, the macroscopic amounts/concentrations of reactants and products remain constant.
- Equal rates do not mean equal concentrations.

# Dynamic equilibrium: what is happening?




A = reactant particles

B = product particles

## Dynamic equilibrium

**A closed system where the forward and reverse reactions continue at the same rate.**

1. **Forward reaction continues**
   A particles collide and convert to B particles.
2. **Reverse reaction continues**
   B particles collide and convert to A particles.
3. **At equilibrium**
   The forward rate = the reverse rate.
4. **Concentrations remain constant**
   The amounts of A and B stay the same macroscopically, although particles are still moving and reacting.

**Equilibrium does not mean the reaction has stopped — it is a dynamic process.**

# Conditions for dynamic equilibrium

- The reaction must be reversible.
- The system must be closed so reactants and products cannot escape or be added unintentionally.
- The forward and reverse reaction rates must become equal.
- Conditions such as temperature should be maintained while the equilibrium state is being considered.
- At equilibrium, observable composition remains constant even though reactions continue.

# How equilibrium is established

- Initially, the forward rate may be high because reactants are abundant.
- As products form, the reverse reaction becomes possible and its rate increases.
- Eventually the forward and reverse rates become equal.
- From that point, dynamic equilibrium is established if the system remains closed under the stated conditions.

# Activity: Dynamic Equilibrium Model

## Teacher action

- Use a reversible reaction and rate model to show forward and reverse processes occurring simultaneously.
- Emphasise equal rates, not equal amounts.

## Student action

- Identify when equilibrium has been established.
- Use the definition and required conditions to justify the decision.

# Rate model at equilibrium

- Before equilibrium: forward rate and reverse rate are different.
- At equilibrium: forward rate = reverse rate.
- After a disturbance, the rates become unequal temporarily.
- The system responds until a new equilibrium is established, where the rates are equal again.

# Common misconceptions about equilibrium

- ‘The reactions stop.’ Incorrect: both directions continue.
- ‘There are equal amounts of reactants and products.’ Incorrect: amounts are constant, but not necessarily equal.
- ‘Any reversible reaction is at equilibrium.’ Incorrect: the required conditions must be met.
- ‘Equilibrium can be maintained in an open container when substances escape.’ Generally incorrect for the defined closed-system equilibrium.

# Independent: Equilibrium Identification Challenge

## Teacher action

• Provide several reversible-reaction scenarios, including open systems and unequal-rate cases.

## Student action

• Decide whether dynamic equilibrium has been established.

• Justify each decision using equal forward/reverse rates and the required conditions.

# Equilibrium Identification: feedback

- Look for a reversible reaction in a closed system.
- Both forward and reverse reactions must still be occurring.
- Their rates must be equal.
- Constant macroscopic composition supports equilibrium, but equal concentrations are not required.

# A12.5.2 Le Chatelier’s principle

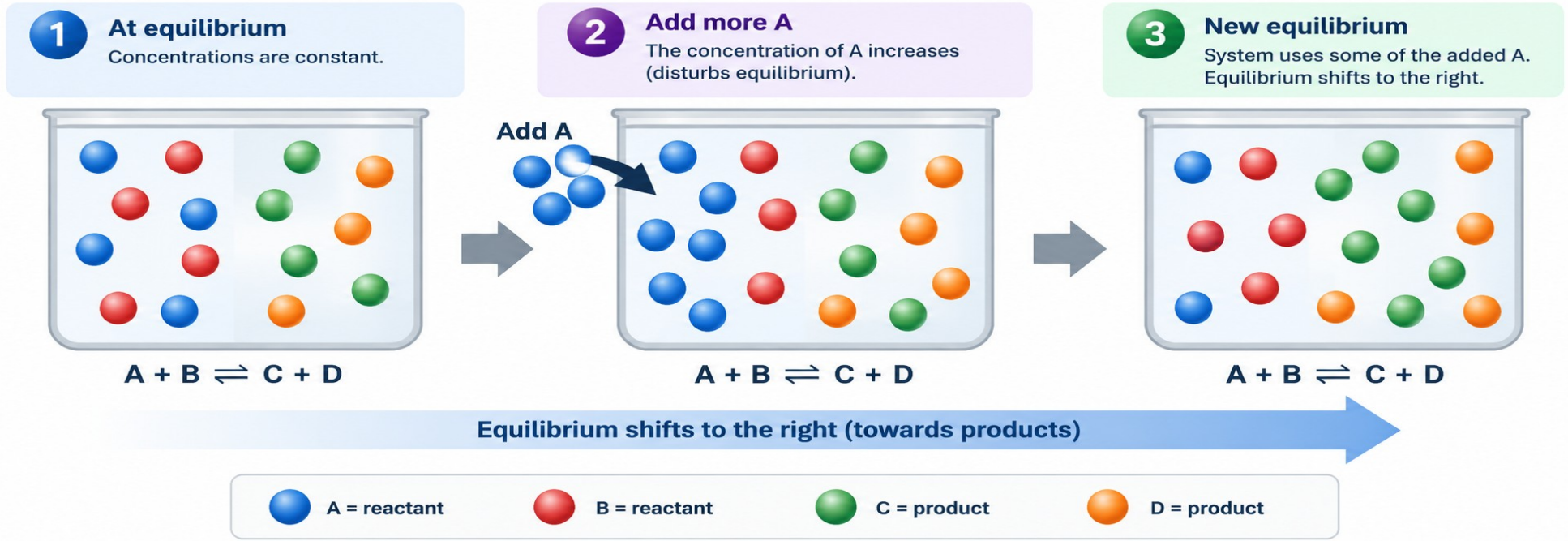
- When a system at equilibrium is subjected to a change in conditions, the position of equilibrium shifts in the direction that opposes the effect of that change.
- Use the principle to predict the direction of shift.
- Then state which side is favoured and how product/reactant amounts change.
- The three required factors here are concentration, temperature and pressure.

# A method for equilibrium predictions

- 1. Identify the change made to the equilibrium system.
- 2. Decide how the system can oppose that change.
- 3. Predict whether equilibrium shifts left or right.
- 4. State which substances are favoured.
- 5. Justify using concentration, heat direction or gas-particle numbers as appropriate.

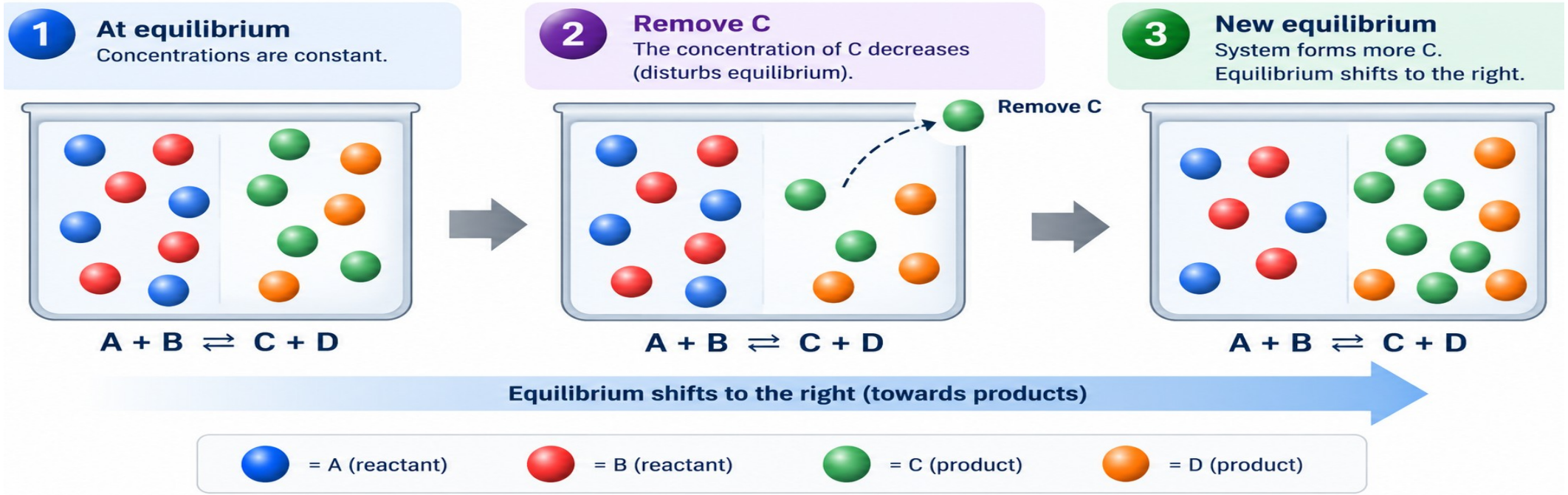
# Changing concentration: adding a reactant

- For A + B ⇌ C + D, increasing the concentration of A disturbs equilibrium.
- The system responds by using some of the added A.
- The equilibrium position shifts toward products: to the right.
- At the new equilibrium, more products are favoured than before the disturbance



# Changing concentration: removing a product

- For A + B ⇌ C + D, removing C reduces its concentration.
- The system responds by forming more C.
- The equilibrium position shifts to the right.
- Removing a reactant would instead favour the reverse direction.



# Concentration Prediction

## Teacher action

• Model adding/removing one reactant or product and explicitly identify the direction that opposes the change.

## Student action

• Predict left/right shift for unfamiliar concentration changes.

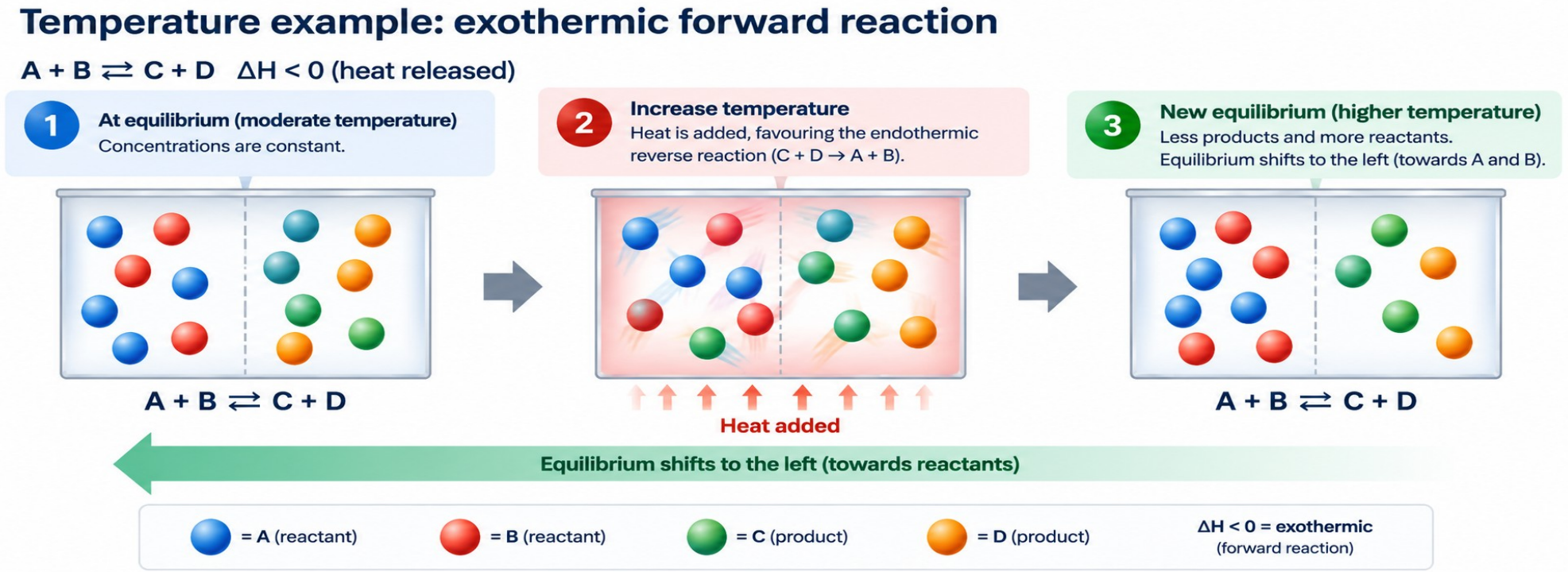
• Explain which substance is being replaced or used up by the response.

# Temperature: treat heat as part of the reaction

- For an exothermic forward reaction, heat can be treated as a product.
- Reactants ⇌ Products + heat.
- Increasing temperature adds heat, so equilibrium shifts in the endothermic direction to use some of it.
- Decreasing temperature favours the exothermic direction.

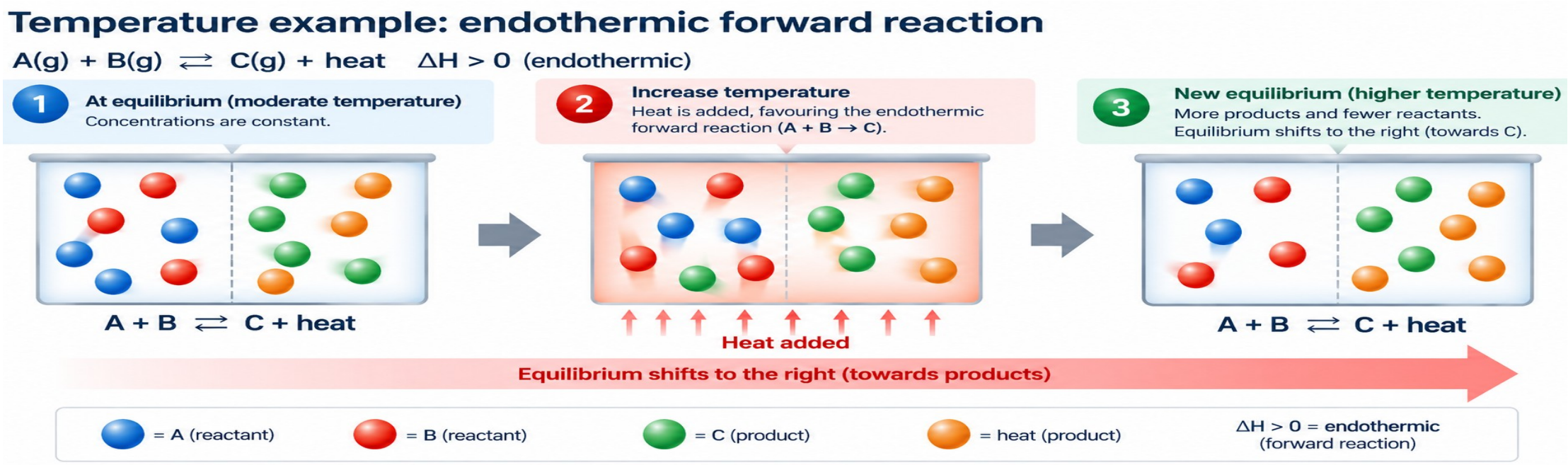
# Temperature example: exothermic forward reaction

- A(g) + B(g) ⇌ C(g) + heat.
- Increase temperature → shift left, favouring the endothermic reverse reaction.
- Decrease temperature → shift right, favouring the exothermic forward reaction.
- Temperature changes the equilibrium position; identify the exothermic/endothermic direction before predicting.



# Temperature example: endothermic forward reaction

- A(g) + B(g) + heat ⇌ C(g).
- Increase temperature → shift right, favouring the endothermic forward reaction.
- Decrease temperature → shift left, favouring the exothermic reverse reaction.
- Do not use ‘left’ or ‘right’ as a memorised rule without knowing the heat direction.



# Temperature Prediction

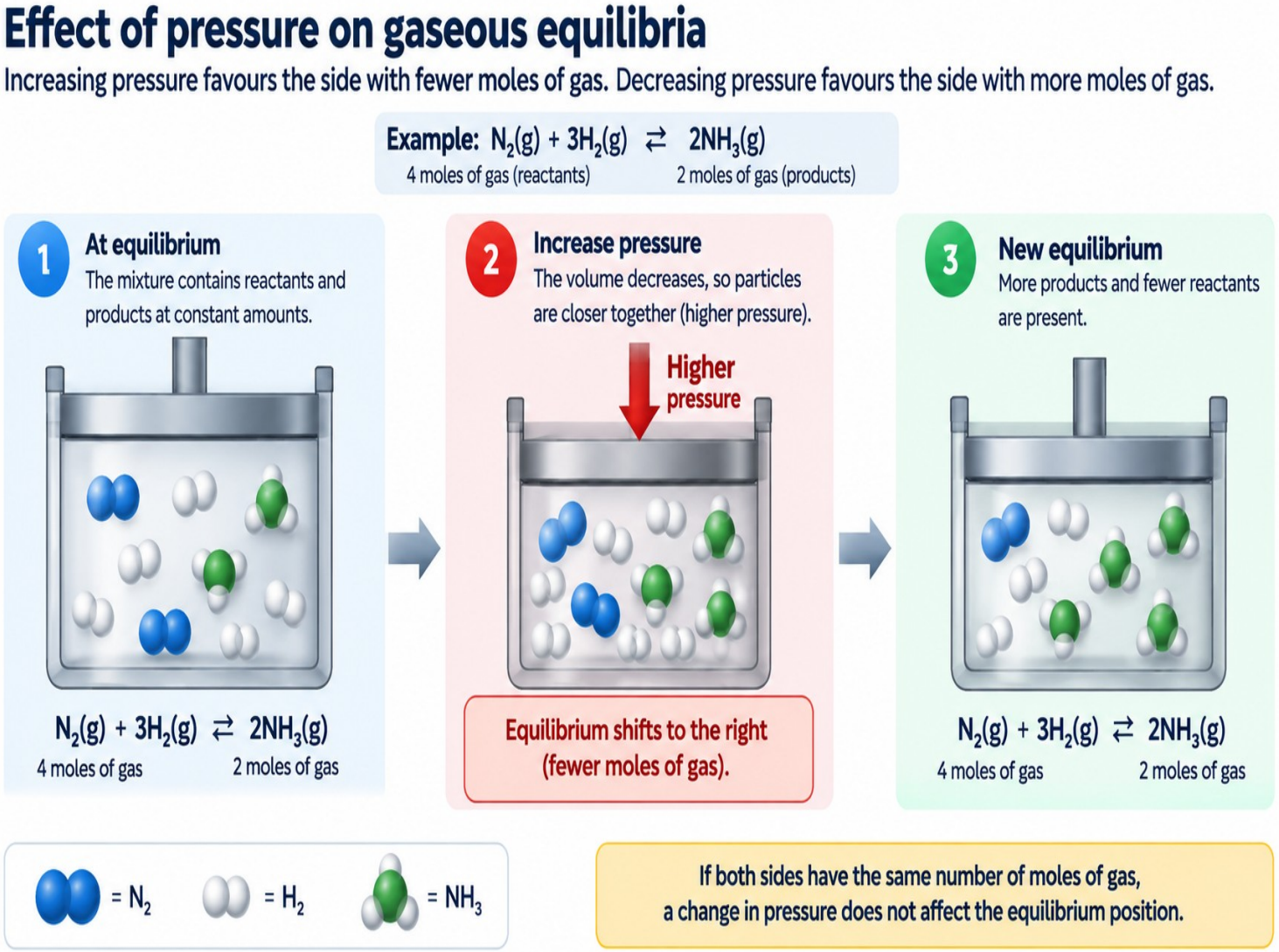
## Teacher action

- Model heat as a reactant or product according to the stated enthalpy direction.

## Student action

- Predict the shift after heating or cooling.
- Justify by identifying the direction that absorbs or releases heat.

# Pressure: only relevant to gaseous equilibria




- Changing pressure affects equilibrium position when gases are involved and the two sides contain different total numbers of gaseous particles/moles.
- Increasing pressure favours the side with fewer moles of gas.
- Decreasing pressure favours the side with more moles of gas.
- If both sides have the same number of gas moles, pressure change does not shift the equilibrium position.

# Pressure example

- $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$.
- Left side: 4 moles of gas in the stoichiometric equation.
- Right side: 2 moles of gas.
- Increasing pressure → shift right, toward fewer gas moles.
- Decreasing pressure → shift left, toward more gas moles.

# Pressure: what to count

- Count gaseous species using their stoichiometric coefficients.
- Do not count solids or liquids when comparing gas moles for a pressure change.
- Example: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$: left has 0 gas moles; right has 1.
- Increasing pressure favours the left side in this model.

# Pressure Prediction

## Teacher action

- Model gas-mole counting before applying Le Chatelier’s principle.

## Student action

- Count gas moles on each side.
- Predict the effect of increasing or decreasing pressure.
- Identify cases where pressure has no effect on equilibrium position.

# Putting the three factors together

- Concentration: shift to oppose addition/removal of a species.
- Temperature: shift in the direction that opposes heating/cooling; identify exothermic and endothermic directions.
- Pressure: for gaseous equilibria, higher pressure favours fewer gas moles and lower pressure favours more gas moles.
- Always justify the prediction rather than stating only ‘left’ or ‘right’.

# Independent: Equilibrium Shift Challenge

## Teacher action

- Give unfamiliar equilibria with concentration, temperature or pressure changes.
- Include at least one no-shift pressure case.

## Student action

- Predict the equilibrium shift independently.
- Justify each prediction using Le Chatelier’s principle.
- State which products/reactants are favoured.

# Unfamiliar examples

- $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, forward reaction exothermic: predict increased pressure and increased temperature.
- $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$: predict increased pressure.
- $A(aq) + B(aq) \rightleftharpoons C(aq)$: predict addition of C.
- For each: state the shift and give the specific reason.

# Unfamiliar examples: answers

- $2SO_2 + O_2 \rightleftharpoons 2SO_3$: increased pressure shifts right (3 gas moles → 2); increased temperature shifts left because the forward reaction is exothermic.
- $H_2 + I_2 \rightleftharpoons 2HI$: 2 gas moles on each side, so changing pressure does not shift the equilibrium position.
- $A + B \rightleftharpoons C$: adding C shifts left, tending to use some of the added product.

# Exam style application

- (a) Define dynamic equilibrium and state one condition required. [3]
- (b) For $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, predict the effect of increasing pressure. Explain. [2]
- (c) The forward reaction is exothermic. Predict the effect of increasing temperature on product yield. Explain. [2]
- (d) Explain the effect of adding a reactant to an equilibrium mixture. [2]

# Exam style answers

- (a) Forward and reverse reactions continue at equal rates (2); in a closed system / appropriate maintained conditions (1).
- (b) Shifts right (1), because the product side has fewer moles of gas: 2 rather than 4 (1).
- (c) Equilibrium shifts in the endothermic/reverse direction (1), so product yield decreases (1).
- (d) Equilibrium shifts in the direction that uses some of the added reactant (1), favouring the opposite side / products if a reactant was added (1).

# Final knowledge check

- Define dynamic equilibrium.
- Why must the system be closed?
- At equilibrium, what is equal?
- Does equal rate mean equal concentration?
- What happens when a reactant concentration is increased?
- How do you predict a temperature effect?
- What does increasing pressure favour in a gaseous equilibrium?